

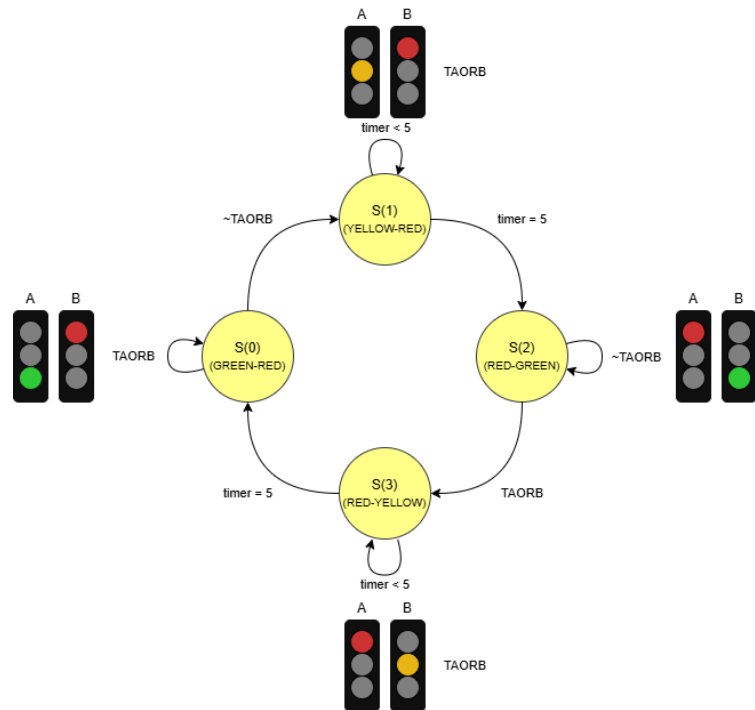
ELE432 Homework 1

Warm-Up – Traffic Light Controller FSM (with Delay)

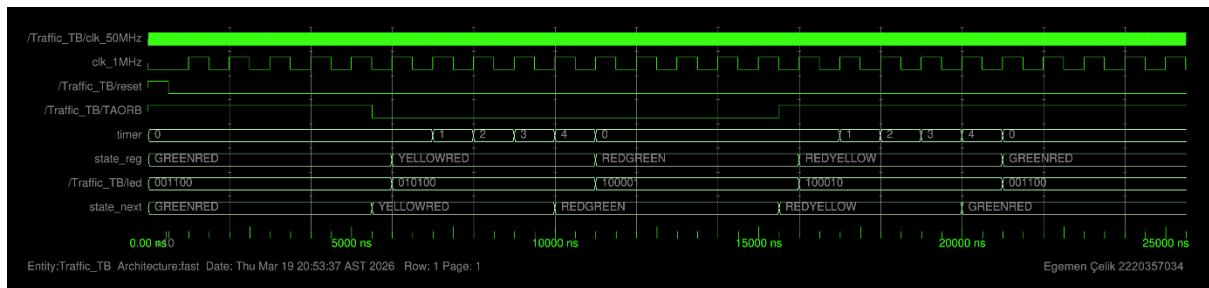
The FSM Behavior given in the homework document didn't state what the state transitions for the TAORB changing while in S(1) or S(3) would be.

I chose to check only one variable on every state as it would be more efficient and the modified behavior would make more sense in the real world.

Modified State Transition Diagram:



Test Bench Waveform:



Files included in the “For grading” folder:

- This document
- Modified State Transition Diagram image
- Test Bench Waveform image
- Transcript Output from Questa
- System Verilog files:
 - 1) Traffic.sv : Logic for the traffic light
 - 2) HalfMicroSecond.sv : 50MHz to 1MHz clock converter
 - 3) TrafficTop.sv : Top level file including both Traffic and HalfMicroSecond instances
 - 4) Traffic_TB.sv : Testbench file for operating under normal conditions.